

JACOB JAMES

Cyber Security Undergraduate

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PROFILE

Cyber Security undergraduate at Manchester Metropolitan University (First Class in Years 1 and 2), with hands-on experience in post-quantum cryptography (including ML-KEM), applied cryptography, network security and vulnerability assessment. Confident working in Python, C and Linux, built through personal projects and a cryptography research internship. Also brings strong communication and customer-service skills from retail work. Looking for a SOC Analyst, Junior Cyber Security, or IT Support role.

EDUCATION

BSc (Hons) Cyber Security, Manchester Metropolitan University 2023 – 2027
• Years 1 & 2 result: First Class

A Levels, The King's CE(A) Sixth Form – IT (Distinction*), Sport (Distinction), Criminology (C) 2021 – 2023

TECHNICAL SKILLS

- **Programming:** Python, C, Java, SQL
- **Cryptography:** hash functions, MACs, digital signatures, hybrid RSA/AES encryption, ML-KEM (Kyber) & post-quantum standards
- **Security tools:** Metasploit, Scapy, vulnerability assessment, network scanning
- **Networking & troubleshooting:** TCP/IP, Linux, Windows, IT troubleshooting, software installation
- **Other:** technical documentation, team leadership, communication, Microsoft Office, Git/GitHub

WORK EXPERIENCE

Cryptography & Cybersecurity Research Intern May – Jul 2026
Manchester Metropolitan University – Research Team

- Researched classical and post-quantum cryptography, including ML-KEM (Kyber) and how quantum computing threatens current encryption standards
- Built a hybrid encryption application in Python using RSA, AES (Fernet) and a custom key management system for secure key storage, rotation and revocation
- Produced technical documentation and presented research findings to the team

Retail Assistant (Seasonal)

Clarks

- Exceeded personal sales targets by more than 5% while assisting customers with product queries and issue resolution

TECHNICAL PROJECTS

Post-Quantum Encrypted Chat Personal Project
Developer | C, OpenSSL, ML-KEM-768

- Built a client/server encrypted chat application using ML-KEM-768 (FIPS 203) for post-quantum key encapsulation and AES-256-GCM for authenticated message encryption
- Implemented HKDF-SHA256 session-key derivation, TCP message framing with partial read/write handling, fresh nonces and tamper detection, testing the cryptographic and networking layers independently before end-to-end validation

Multi-Site Enterprise Network Design Personal Project
Network Designer | Cisco Packet Tracer

- Designed a 50+ device enterprise network across five sites – two Manchester offices, a wireless-only Manchester factory, and two London offices – using a structured CIDR /24 addressing scheme with VLAN segmentation per department
- Configured RIP v2 dynamic routing, DHCP, DNS and SSH for secure device management, then verified the design with ping/tracert testing and a simulated router failure to confirm automatic rerouting

Homelab Server Personal Project
Linux (Ubuntu Server), Docker, Python/Bash

- Built and maintain an Ubuntu Server homelab, writing custom Python and Bash scripts for system monitoring, automated backups and user management
- Self-hosted a stack of seven services – Nginx, Traefik, Docker (via Portainer), Prometheus, Grafana, Uptime Kuma and Bind9 – all tied together on a single dashboard

Phish Trap – Phishing Email Detector Personal Project
Product Owner | Python | Team of 6

- Led development of a phishing detection tool using keyword and rule-based analysis, achieving 98%+ detection accuracy in internal testing

Metasploit Vulnerability Security Audit Personal Project
Team Lead | Metasploit Framework

- Led a vulnerability assessment using Metasploit and wrote up the findings in a report focused on fixes, keeping testing ethical throughout

INTERESTS

Capture The Flag (CTF) participant • Home lab building • Badminton • Hiking